

Borel–Cantelli, Sequence Convergence, and Radon–Nikodym

Topic 3

Contents and the questions they answer

1. Borel–Cantelli: recurrence of events
2. Random-variable convergence: errors and subsequences
3. Convergence of expectations and densities
4. Radon–Nikodym and change of measure
5. Sampling applications

Event sequences: when do events occur infinitely often?

Random-variable sequences: apply this to errors $\{|X_n - X| > \varepsilon\}$.

Expectations: what extra condition permits a limit under the integral?

Change of measure: establish density formulas, then use them to improve sampling.

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is the probability space. Topic 1 supplies MCT, Fatou and DCT. Topic 2 supplies laws, expectations and tail integrals.

Infinitely often and eventually always

For events $A_n \in \mathcal{F}$

Definition 1 (Infinitely often)

Theorem 2 (First Borel–Cantelli lemma)

A finite count on every path can have infinite expectation

On $(\Omega, \mathcal{F}, \mathbb{P}) = ((0, 1), \mathcal{B}((0, 1)), \lambda)$, let $U(\omega) = \omega$ and $A_n = \{U \leq 1/n\}$.

Definition 3 (Mutual independence)

Theorem 4 (Second Borel–Cantelli lemma)

Recurrence of a fixed word in coin tosses

Example 5 (Independent fair tosses (ξ_n)) Fix a binary word w of length ℓ . We want infinitely many copies of w .

Almost-sure convergence and error events

Throughout this section, X_n and X are real-valued random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 6 (Almost-sure convergence)

Summable deviation probabilities

Corollary 7 (Summable deviations)

Proof

Example: a summable moment bound

Finite first moment: $X_n/n \rightarrow 0$ almost surely

Corollary 8 (Sublinear observations)

Proof

Tail integrals give a summability criterion

Tail-series comparison: intervals of width $h > 0$

For $Y \geq 0$, write $T(t) = \mathbb{P}(Y > t)$, a decreasing function.

Infinite first moment: $\limsup |X_n|/n = \infty$

Theorem 9 (Independent, identically distributed (X_n))

Proof

Almost-sure convergence implies convergence in probability

Definition 10 (Convergence in probability)

Proposition 11 (Almost surely implies in probability)

Proof

The typewriter sequence

On $([0, 1], \mathcal{B}([0, 1]), \lambda)$, take $U(\omega) = \omega$ and define $X_{2^m+k} = \mathbf{1}_{\{k2^{-m} \leq U < (k+1)2^{-m}\}}$ for $0 \leq k < 2^m$.

Dyadic intervals and their indices

The subsequence criterion

Theorem 12 (Subsequence criterion)

Remark: one convergent subsequence is insufficient

Continuous mappings

Theorem 13 (Continuity at the random limit)

Proof

Threshold probabilities and discontinuities

Assume $X_n \xrightarrow{\mathbb{P}} X$.

Application to a threshold

Convergence of expectations: the missing integrable bound

Counterexample 14 (The shrinking spike) On $([0, 1], \mathcal{B}([0, 1]), \lambda)$, let $X_n = n\mathbf{1}_{[0, 1/n]}$.

Recall: Fatou and dominated convergence

Dominated convergence in probability

Theorem 15 (A common integrable bound)

Proof: the limit is integrable

Dominated convergence in probability: proof

Convergence of the integral errors

A further subsequence and DCT

Scheffé's lemma

Theorem 16 (Nonnegative Scheffé lemma)

Probability densities

Scheffé's lemma: proof

Proof: the common part

Density convergence and event probabilities

On $(S, \mathcal{M}, \mu)$, let f_n, f be finite nonnegative measurable functions with $\int f_n d\mu = \int f d\mu = 1$, and $f_n \rightarrow f$ a.e.

Scheffé gives convergence in L^1

Positive and negative parts

An exact calculation of the probability error

On $((0, 1), \mathcal{B}((0, 1)), \lambda)$, set $f = 1$ and $f_n = (1 + 1/n)\mathbf{1}_{(0, a_n)}$, where $a_n = n/(n + 1)$.

Normalization and pointwise convergence

The absolute error

Absolute continuity and a probability density

So far, the densities were given. For two probabilities $\mathbb{P}$ and $\mathbb{Q}$ on $(\Omega, \mathcal{F})$, when can we write $\mathbb{Q}(A) = \int_A L \, d\mathbb{P}$?

Definition 17 (Absolute continuity)

A probability defined by a density

Countable additivity

Theorem 18 (Radon–Nikodym)

Uniqueness of the Radon–Nikodym derivative

Uniqueness: the event where one density is larger

Absolute continuity controls small-event probabilities

Corollary 19 (Fixed probabilities $\mathbb{Q} \ll \mathbb{P}$)

Proof: truncate the density

Convergence under a fixed change of measure

Convergence in probability

A change of probability on two outcomes

Example 20 (An explicit Radon–Nikodym derivative) On $\Omega = \{\omega_1, \omega_2\}$ with $\mathcal{F} = 2^\Omega$, let

$$\mathbb{P}(\{\omega_1\}) = p, \quad \mathbb{P}(\{\omega_2\}) = 1 - p, \quad 0 < p < 1.$$

To assign probabilities q and $1 - q$, where $0 \leq q \leq 1$

Change of measure in expectations

Proposition 21 (Expectation conversion)

Proof: simple functions

Change of measure in expectations: proof

Nonnegative measurable functions

Chain rule for Radon–Nikodym derivatives

Proposition 22 (Chain rule)

Proof

The reciprocal density and reverse expectation formula

Suppose $L = dQ/dP > 0$ $\mathbb{P}$ -a.s.

Mutual absolute continuity

The reciprocal density

Expectation under the original probability

An unbiased estimator can have infinite variance

Example 23 (An integrable singularity) Let $0 < \alpha < 1$, $h(u) = u^{-\alpha} e^{-u}$, and $I = \int_0^1 h(u) \, du$.

The uniform-sampling estimator

When the variance is infinite

Variance and the error of a sample average

Mean squared error of the sample average

Importance sampling: choice of the density

On $\Omega = (0, 1)$ with $\mathcal{F} = \mathcal{B}((0, 1))$, let $\mathbb{P} = \lambda$ and $h(x) = x^{-\alpha} e^{-x}$, where $0 < \alpha < 1$. The target is $I = \int h \, d\mathbb{P}$.

Choice of the sampling density

Importance sampling: cancellation of the singularity

The bounded weighted estimator

Sampling the chosen density with a uniform variable

The sampling law has density $q(x) = (1 - \alpha)x^{-\alpha}$ on $(0, 1)$.

CDF and inverse transform

A sample from q

The value returned by one sample

Gaussian change of measure

Fix $a \in \mathbb{R}$ and let $X \sim N(0, 1)$ under $\mathbb{P}$. We want this same X to have law $N(a, 1)$ under $\mathbb{Q}$. Write $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$.

The density ratio

Normalization

Composition of Gaussian shifts

A second shift centered at the current mean

Gaussian rare-event sampling

Let $X \sim N(0, 1)$ under $\mathbb{P}$ and $p_b = \mathbb{P}(X > b)$, where $b > 0$.

Sampling near the threshold

An unbiased estimator of the original tail probability

Gaussian rare-event sampling: exact second moment

The second moment under $\mathbb{Q}$

Completing the square

Gaussian rare-event sampling: variance improvement

A direct bound for comparison with Bernoulli sampling

Supplement: Fatou and a monotone derivative

Example 24 (When only an inequality is justified) Let $f : [0, 1] \rightarrow \mathbb{R}$ be nondecreasing and differentiable a.e. Set $f' = 0$ where the derivative does not exist.

Nonnegative difference quotients

Supplement: the integral bound and strict inequality

Integral of the difference quotient

Fatou and strict inequality